Introduction to Software Engineering

Project Idea & Team Contract

The student team is required to complete the **Project Idea & Team Contract** documentation for the assigned course project, following the attached template.

Software Engineering Department
Faculty of Information and Technology
University of Science

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Project Idea & Team Contract

Objectives

This document focus on the following topics:

- ✓ Describing the project idea at a high level so the teaching staff can approve it
- ✓ Agreeing on how the team will work together (team contract)
- ✓ PA0 is not graded, but it is required before the team starts PA1

1 Team Information

Item	Value
Team ID	6 (course group number; check T06 format for Moodle)
Team name	404NotFound
Team lead	Nguyễn Huy Hoàng (24127378)
Communication channel	General | 404NotFound

Student ID	Full name	Email	Main role	Backup role
24127378	Nguyễn Huy Hoàng	nhhoang2435@clc.fitus.edu.vn	Team lead / PM; Backend contributor	Business analyst; QA
24127085	Võ Quang Minh	vqminh2440@clc.fitus.edu.vn	Business analyst; UI/UX; Frontend	Team lead / PM; Developer
24127499	Lê Hoàng Phúc	lhphuc2417@clc.fitus.edu.vn	Backend; Database; CI	Tester; Frontend
24127568	Nguyễn Cao Trí	nctri2407@clc.fitus.edu.vn	Tester / QA; Integration (proposed, NOT confirmed)	Developer; Designer (proposed, NOT confirmed)

PA0 role allocation is PROPOSED, not confirmed. All four members are expected to contribute technical work (implementation, tests, integration, and reviews) as well as their primary coordination or documentation duties. Nguyễn Cao Trí has not yet responded; his assignment below is tentative and does not imply consent. Confirm roles with each member and the TA before final submission.

2 Project Idea Overview

Proposal for discussion and TA approval. Assumes a small cafe or retail store with part-time staff; no claim is made about any real company.

2.1 Project Name and One-Sentence Pitch

ShiftManager (working title) - a lightweight web application that helps small businesses schedule employee shifts and safely exchange shifts with co-worker consent and final manager approval. The name is already used by other products; the team will confirm its final project name before submission.

2.2 Problem and Motivation

In a hypothetical cafe, managers prepare shift rosters in spreadsheets and employees coordinate changes via chat. This can make it difficult to know which roster is current, whether a colleague consented, or whether a proposed swap would overlap another assigned shift. ShiftManager centralises a simple workflow: publish shifts, request a two-way shift swap, obtain the other employee's response, validate the schedule, and record a manager's decision. This scenario is a proposed problem to validate with the TA or potential users, not a verified claim about a specific business.

2.3 Target Users

User group	What they need from the system
Employee	Sign in, view own assigned shifts, request an exchange of two future shifts, accept/decline incoming requests, and see the decision.
Manager	Create and assign shifts; view the team schedule; approve/reject proposed swaps with conflict checks.
Small business owner (stakeholder)	Stakeholder only (no separate login role): seeks an accurate, current team roster; the system does not promise a full audit trail in the Must-have scope.

2.4 Main Features

Eight candidate features; F01-F06 are Must-have for PA5. F07-F08 are optional and excluded from the committed MVP.

#	Feature	Short description	Must-have?
F01	Sign-in & role access	Use pre-created accounts; restrict employee and manager actions.	Yes
F02	Shift assignment	Manager creates future shifts and assigns an employee, rejecting overlapping assignments.	Yes
F03	Schedule views	Employees view their own roster and minimal details of eligible co-worker shifts; managers view the team roster.	Yes
F04	Swap request	Employee requests a two-way exchange between their future shift and an eligible future co-worker shift.	Yes
F05	Co-worker response	Invited colleague accepts or declines; no swap happens on acceptance alone.	Yes
F06	Manager decision	Manager approves/rejects; re-check ownership, start time and both schedules; on approval, swap owners atomically.	Yes
F07	Decision history	Optional detailed timestamped history of responses and managerial decisions; the core MVP still shows the current request status.	No
F08	Email notifications	Send alerts when a request receives a response or decision.	No

2.5 Similar Products

Similar products: Deputy (https://www.deputy.com/features/shift-swapping), Homebase (https://www.joinhomebase.com/employee-scheduling/shift-swapping) and Shift Manager (https://shift-mgr.com/). They support scheduling and/or shift swaps. Our academic prototype is intentionally smaller: one workplace, two roles, two-way swapping and manager approval, without payroll or external integrations. The working name is already in use elsewhere.

2.6 Platform and Technology Ideas

Responsive web application. Proposed stack, subject to team approval: React/JavaScript frontend; Node.js/Express REST backend with role-based access; Supabase for relational database and transactions; Git/GitHub for reviews; GitHub Actions for CI. Use fictional business data and seeded demo accounts. Verify team skills and deployment options before PA1.

2.7 Feasibility, Scope and Open Questions

- Feasibility: Six Must-have features may be practical in the main ~3-week coding window (weeks 8-10) if the team limits the app to one workplace, seeded demo accounts, a simple roster and two-way swaps. Four students are registered, but Tri's participation is not yet confirmed. Proposed business rules: each requested shift is in the future, owned by a different participating employee and not involved in another unresolved swap; the invited colleague must consent, followed by final manager approval. On approval, re-check ownership, shift start times, and both employees' schedules, then update both assignments within a single PostgreSQL transaction. Adjacent shifts do not overlap. Employees may view limited information about eligible co-worker shifts solely to propose a swap.
- Out of scope: clock-in/out, payroll, leave management, automatic scheduling, one-way giveaways, multiple branches, push notifications and external HR integrations.
- Risks: unknown availability of one member (Tri); new React/Express/PostgreSQL stack; incomplete conflict rules; scope growth; late tests and deployment. Mitigation: request TA guidance, assess skills early, write concrete swap rules/tests, freeze the Must-have list, and deploy a minimum end-to-end flow early.
- Questions for TA: Is a fictional small-business scenario acceptable? Are seeded demo accounts allowed? How should an unreachable member sign PA0?

3 Team Contract

3.1 Goals and Expectations

Proposed shared goal: deliver all Must-have features, consistent PA1-PA5 documentation and a working demonstration, aiming for Good or better on the project rubric. Proposed effort: each member contributes about 5-7 hours/week outside class, with additional mutually agreed work near deadlines. All four members are expected to participate in technical tasks; leadership or documentation work does not exempt anyone from coding, testing and reviews. These are proposals until every member confirms their availability and expectations.

3.2 Communication

- ✓ Main discussion channel: Microsoft Teams. Links: General | 404NotFound
- ✓ Proposed response time: within 24 hours on weekdays; mark urgent blockers and call an active member when necessary.
- ✓ Planned record: GitHub repository and Issues/Project board for tasks/decisions, plus /docs/management/weekly for reports. Repository/board URLs: https://github.com/gloryh241106/ShiftManager

3.3 Meetings

- ✓ Weekly status meeting with TA: 19:00, Thursday, Online
- ✓ Internal team meeting: 19:00, Friday; shorter check-in before every PA deadline.
- ✓ Proposed attendance rule: notify the team in advance; record absences and resulting actions honestly in the weekly status report.
- ✓ Weekly report deadline: 19:00, Sunday

3.4 Roles and Responsibilities

Proposed responsibilities (subject to personal confirmation): Hoang - team lead/PM, task tracking, backend API/business logic, test coordination and PA integration; Minh - business analysis, UI/UX design and React frontend implementation; Phuc - Node.js/Express backend, PostgreSQL schema, integration and GitHub Actions CI; Tri - testing/QA, test automation and frontend/backend integration support (TENTATIVE: not yet contacted). All four take part in implementation, test design/execution, Git commits and pull-request reviews, with at least one

reviewer other than the author. Backup coverage: PM by Minh, BA by Hoang, Designer by Tri (tentative; otherwise Minh/Phuc), Developer by Hoang/Minh, Tester by Phuc. The team must ratify the division and reassign work as needed with TA guidance; none of this represents Tri's acceptance.

3.5 Work Standards

- ✓ Proposed internal deadline: each contributor submits material at least 48 hours before the official PA deadline for peer review.
- ✓ Quality: each PA section and pull request reviewed by at least one other team member; keep requirements, designs, tests and commits consistent.
- ✓ Tools: track owner and due date for every task; use branches, reviewed pull requests, and a protected main branch once set up.

3.6 Use of AI Tools

Proposed permitted use: ChatGPT or similar tools for brainstorming, wording, code explanation and test ideas, never as unreviewed final work. Every member must check results against the requirements and be able to explain the submitted material. The person using AI records significant usage; Hoang consolidates the log for each PA (proposal). Do not enter confidential data.

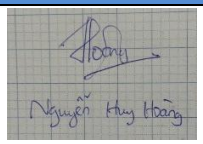
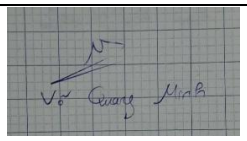
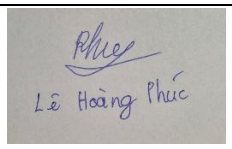
3.7 Decision Making

Proposed policy: discuss major scope/design choices first; if no consensus, use one vote per member who is participating. For a tie or unresolved issue, seek TA guidance instead of silently imposing a decision. Record decisions in the task board.

3.8 Handling Conflicts and Missed Work

- ✓ *Step 1: talk directly with the member and agree on a fix and a new date*
- ✓ *Step 2: discuss in a team meeting and record it in the weekly status report*
- ✓ *Step 3: inform the TA; the issue is taken into account in the individual contribution factor*
- ✓ Step 4 (proposed): for unreachable members, keep evidence of contact attempts (course email and Messenger), do not record consent or signatures on their behalf, and ask the TA how to proceed; reassign only documented at-risk tasks.

3.9 Signatures

Student ID	Full name	Signature	Date
24127378	Nguyễn Huy Hoàng		13/10/2026
24127085	Võ Quang Minh		13/10/2026
24127499	Lê Hoàng Phúc		13/10/2026
24127568	Nguyễn Cao Trí	[NOT SIGNED]	

4 AI Usage Log

AI-assisted brainstorming, fact-checking of a reused product name, and revision of the scope, technology proposal and role allocation are disclosed below. Team verification and approval are still pending; update the log with each real use and review step before final submission.

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Huy Hoàng	ChatGPT	Brainstorm a business software idea; draft and revise PA0; assess a duplicated project name; propose React/Express/PostgreSQL and roles.	AI draft checked against the supplied HCMUS template and publicly described similar products; missing team decisions, signatures and human review remain explicitly marked pending. Members must verify and revise before submission.